Trade-off Between Trade Efficiency and Immediacy: Evidence from Switching to NASDAQ

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Abstract

The trade-off between execution immediacy and price improvement has explained the choice of traders between dark or lit markets. We show that this choice is altered by the same trade-off resulted from trading in dealer versus auction markets. We find that the switch to NASDAQ increases execution cost, partly due to the higher preferencing agreement in NASDAQ, and in response, uninformed trades, which are more sensitive to cost changes, move to dark markets. Our results show that the informational efficiency improves in the lit market at the expense of lower liquidity, when a firm moves from NYSE to NASDAQ.

JEL classification: G14, G32, G41

Keywords: NYSE, NASDAQ, listing, dark trading

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1. Introduction

Tse and Devos (2004) show that moving from NYSE to NASDAQ increases the trading cost. In this paper, we find the same negative effect on quoted bid-ask spreads when a firm moves to NASDAQ, and show that the increase in the execution cost alters the price-time equilibrium between dark and lit markets. Following the move to NASDAQ, uninformed trades, who are more sensitive to trade cost, are siphoned from lit market to dark pools to alleviate the rise in the execution cost at the expense of lower trade immediacy. We find that the shift of trades to dark markets increases dark liquidity, measured by relative effective spread. Further, we show that the relative realized spread decreases in the lit market, indicating an improve in price discovery.

Cohen, Maier, Schwartz, and Whitcomb (1981) explain one of the several price-time trade-offs before the traders when a trader decides to use a market order versus a limit order. If traders have to trade fast on a private information, they are expected to use market orders, while if they are patient and care more about price improvement and low cost trade, they will choose limit orders. The trade-off between price and time also occurs in the choice of a trading venue between auction and dealer markets. Aggarwal and Angel (1997) argue that firms face a trade-off between the lower cost of an auction market and the marketing advantage of a dealer market. Auction markets, such as NYSE, have a lower execution cost and higher price improvement as compared to dealer markets, such as NASDAQ. In dealer markets, though, the presence of dealers improves immediacy of order execution, as dealers are almost always there to take a position. One of the major reasons for the high trading cost in dealer markets is the prevalence of preferencing. Preferencing agreements between market centers and brokers who route customers' orders allow market centers to obtain order flow regardless of their quotes and this may result in a lack of competition in the market, and even though alternative market centers, whether NYSE or NASDAQ, can compete for order flow by posting aggressive quotes, they obtain significant portion of their orders through preferencing. These agreements are much more prevalent in NASDAQ due

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to its dealer market nature. Weston (2000) argues that the 1997 NASDAQ reforms have reduced NASDAQ dealers' rents, however, He, Odders-White, and Ready (2006) document a persisting high preferencing for NASDAQ stocks and find that dealers in NASDAQ capture additional profits of up to 55% of the effective spread.

In addition to the aforementioned trade-off mechanisms, Menkveld, Yueshen, and Zhu (2017) introduce another price-time trade-off by arguing that after investors receive urgency or cost shocks in real-time, they move between low-cost, low-immediacy venues (dark venues) and high-cost, high-immediacy venues (exchanges). This pecking order theory and its predecessors, such as Zhu (2014), explain how the choice between price improvement and informational advantage segments the market between dark and lit. Zhu (2014) argues that exchanges are more attractive to informed traders, and dark pools are more attractive to uninformed traders, referring in part to the higher sensitivity of uninformed traders to trade cost. In the arrival of a cost shock, uninformed traders are more motivated to move from lit to dark, resulting in price discovery improvement as well as liquidity deterioration in exchange (lit) markets. In this paper, we show how the change in the dealer-auction market equilibrium can affect the dark-lit market equilibrium, and how liquidity and price discovery are affected in the new equilibrium. When a firm switches from NYSE (an auction market) to NASDAQ (a dealer market), it creates a cost shock to the traders, both informed and uninformed. More sensitive to the cost shock, the uninformed traders decide to leave the exchange and move to dark venues to relieve the cost shock. Thus, the liquidity in the lit market decreases while informational efficiency and price discovery improve.

Establishing the mechanism, we take a step back and discuss whether this mechanism can support the decision of the managers to switch from NYSE to NASDAQ, along with other reasons expressed in the previous papers, such as Dang, Michayluk, and Pham (2018) and Tse and Devos (2004). We test the classical explanations stated by the firms' managers in their switch announcements. Our tests show that while the market reaction to switch announcement is insignificant, liquidity and trade volume are negatively affected by the switch,

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with relatively large magnitude. We find insignificant increase in coverage of analysts and insignificant decrease in institutional investors' concentration and capital raise. Additionally, we find no systematic industry effect that could motivate a decision to move on the basis of joining a "club" of, say, tech firms on NASDAQ. We also examine CEO restlessness as a behavioral reason for the switch; however, we find no significant results. Therefore, we conjecture that managers choose to switch to NASDAQ to increase price discovery at the expense of execution cost, following our proposed price-time trade-offs mechanism.

Our paper makes several important contributions to the optimal trading choice and dark pools trading literature. It is in line with Jiang, McInish, and Upson (2012), who find that the price discovery on exchanges improves in fragmented markets because uninformed traders are able to route their order flow to off-exchange trading venues, leaving a larger proportion of informed traders at the exchange.

The rest of the paper follows this structure: In Section 2, we review the literature on dark pools and on comparison between NYSE and NASDAQ, and propose our hypotheses. Section 3 explains the data and variable. Section 4 presents our analyses and results. In section 5, we examine several rival explanations for switching to NASDAQ. Finally, Section 6 concludes.

2. Dark Trades, Listing Switch, and Hypotheses Development

2.1. Dark vs. Lit Trades

Dark pools are one of the forms of ATS (Alternative Trading Systems) markets, in which trade transparency is less than in exchange markets. Dark and lit (exchange) trades differ in their levels of pre-trade transparency, but not post-trade transparency (Comerton-Forde and Putniņš (2015)). Pre-trade transparency is the degree to which information is available to market participants about buying or selling interest, including quotes to buy or sell and

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the volumes available at the quotes. Post-trade transparency is the information about trades after they execute. A considerable fraction of trade volume belongs to dark pools. 14% of the consolidated volume trades in 19 dark pools in the US. In Europe, 16 dark markets account for approximately 4.5% of the volume, and in Canada, dark pools represent 2% of the volume (Buti, Rindi, and Werner (2017)). The big development in dark pools came in 2007 when the SEC passed Regulation NMS (National Market System), which allowed investors to bypass public exchanges to gain price improvements. The goal was to attract a number of new players to the market, and, as a result, a large number of dark pools were created over the next ten years. It is fair to say that there are three main features differentiate lit markets from dark venues, which are partially exempt from the NMS requirements. First, compared with lit markets, dark venues currently are not subject to NMS fair-access requirements and thus can prohibit or limit clients from accessing their services. Second, dark venues provide limited or no pre-trade transparency in that their best-priced bids and offers are not required for inclusion in the publicly distributed consolidated quotation data. Finally, executions on dark venues occur at finer price increments than on lit markets.

There is a considerable body of theoretical and empirical research on the choice of traders to trade in dark markets or exchanges. Dark pools are required to match orders at prices within the National Best Bid and Offer (NBBO), while exchange orders can be submitted to execute immediately even if the resulting execution prices fall outside of the prevailing NBBO. While this approach in dark pools can lead to lower bid-ask spread costs, if no match within the NBBO exists, then no trade will occur. Menkveld et al. (2017) proposes the pecking order hypothesis which predicts that as investors' trading needs become more urgent, they move from low-cost, low-immediacy venues to high-cost, high-immediacy venues. More concretely, recent theories of dark pools mentioned above predict that dark venues (low-cost, low-immediacy) are at the top of the pecking order, whereas lit venues are at the bottom (high-cost, high-immediacy), because dark venues provide potential price improvement but lack execution certainty. In another seminal theory papers, Zhu (2014) proposes a model

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of venue choice that allows both informed and uninformed traders to trade on exchanges and dark pools. Based on the model, since informed traders trade in the same direction and crowd one side of the market, they face a higher execution risk (larger implied bid-ask spread, indicating a higher execution cost) in a dark pool than uninformed traders face. Therefore, exchanges will be relatively attractive to informed traders, and dark pools will be relatively attractive to uninformed traders. Consistently, Jiang et al. (2012) and Foley and Putniņš (2016) also show that the traders in dark pools are mainly uninformed. Jiang et al. (2012) find that the price discovery on exchanges improves in fragmented markets because uninformed traders are able to route their order flow to off-exchange trading venues, leaving a larger proportion of informed traders at the exchange.

2.2. Switching to NASDAQ and Its Effect on Dark Trades

The choice of a stock listing market is a relatively permanent one. Once firms choose to IPO on one market, they generally remain listed on that market indefinitely. All the same, if a switch occurred, it was exclusively from NASDAQ to NYSE. However, Aeroflex Co. was the first company to change its listing to NASDAQ in March 2000. The economic effects of this change have been explored with mixed results. Kalay and Portniaguina (2001) found a significant positive abnormal return, and a reduce in the daily bid-ask spread and the trading volume; On the contrary, Pruitt, Van Ness, and Van Ness (2002) found no improvement in trading or quoting behavior, but a deterioration in key metrics. These conflicting results might explain the dearth of follow-on switches in the following years. Nevertheless, in 2008, the market recorded a jump from essentially none to almost 50 firms opting out of NYSE and switching to NASDAQ. In 2009, despite the market-wide drought in listings in the aftermath of the crisis, firms kept on switching at an unprecedented number, from NYSE to NASDAQ. Switches continued in the following years and, since 2010, the numbers of switches between the two markets in the two directions have been essentially the same (excluding ETFs and ETNs). Figure (1) illustrates the distribution of switches over the years for both directions.

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This phenomenon is economically significant in light of the widely publicized (NASDAQ (2017b)) milestone, reached by NASDAQ, which, in September 2017, announced to have passed the 1 trillion dollars in market cap, snapped from NYSE.

[Insert Figure 1 here]

Although the weight of literature is larger on the switches from NASDAQ to NYSE, there are very few papers that find benefits in switching to NASDAQ, and try to explain the rationale. Among them, Cheon, Christensen, and Smith Bamber (2001) show that abnormal returns around the earning announcements are larger for firms listed on NASDAQ compared to those on NYSE, and Masulis and Shivakumar (2002) compare the speed of price adjustments to seasoned equity offering announcements by NYSE/AMEX and NASDAQ, and show that NASDAQ is faster. Execution times for orders in NASDAQ stocks are lower than for orders in NYSE-listed stocks, also consistent with Boehmer (2005)'s results. Speed is presumably an important aspect of execution quality to traders with short investment horizons and to investors who base trading decisions on time-sensitive information (He et al. (2006)). The most recent papers that study the aspects of switch to NASDAQ find an increase in execution cost upon the switch. Tse and Devos (2004) show that firms that move to NASDAQ increase investor recognition and volume at the expense of an increase in trading costs. Dang et al. (2018) find the move to the NASDAQ increases trading costs, while improves visibility and attracts more liquidity providers in the long term.

Few papers have investigated the level of dark trades in the two groups of NYSE-listed stocks and the NASDAQ-listed ones. Buti, Rindi, and Werner (2011) show that dark pool activity is generally higher for NASDAQ-listed stocks than NYSE-listed ones. They attribute this difference in dark pool activity to the type of stocks listed and mainly traded in each of these two exchanges. They show that there is an association between dark pool activity and firm size (or stock capitalization).

The above arguments lead us to propose the following hypotheses: When a firm moves

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to NASDAQ, the preferencing friction increases and therefore the cost of trade (in the entire market, including both lit and dark) increases. Accordingly, we expect that upon the switch the quoted bid-ask spread widens:

Hypothesis 1) Following the switch to NASDAQ, execution cost (quoted bid-ask spread) increases.

Since the cost of trade is increased, uninformed traders, who are more sensitive to cost changes, decide to move to a less expensive markets, namely dark pools:

Hypothesis 2) Following the switch to NASDAQ, a significant volume moves from lit markets to dark pools.

The asymmetry of sensitivity with respect to the execution cost between informed and uninformed traders segments the market and increases the relative number of informed traders in the lit market. Since more informed traders have moved to the dark markets, its liquidity improves. We decompose the quoted bud-ask spread to effective and realized spreads to have a more precise measure for the hidden liquidity (effective spread) in the dark pools and price efficiency (realized spread) in the lit markets:

Hypothesis 3) After switching to NASDAQ, the liquidity (effective spread) in the dark market improves, the liquidity in the lit market deteriorates, and the informational efficiency (realized spread) in the lit market improves.

In the rest of the paper, we test these hypotheses.

3. Data and Variables

3.1. Data Sources

We obtained the data on the switching firms from NASDAQ Economic and Statistical Research Department. The data set includes the full list of issuers who switched to and from NASDAQ in the period 2009-2017. Each switching issuer is identified by name, ticker, date of announcement, date of effective switch, and market cap at the time of the switch. The

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sample excludes ETFs, ETNs, and includes only corporate switchers. Since the financial crisis period has profoundly affected the pricing and trading behavior of all securities, we narrow the sample to switches announced in 2009 and later. Our final sample consists of 110 firms that switched from NYSE to NASDAQ between 2009 and 2017, and 72 firms that switched from NASDAQ to NYSE in the same period. Interestingly, there are two firms that switched twice: first from NASDAQ to NYSE and after a few years back to NASDAQ. While anecdotally interesting, this phenomenon is isolated and does not affect our statistical results.

We collect the total volume of lit and dark trades from CRSP, reported in *vol* variable. Our source of data for ATSs is TAQ database. Since the late 1970s, all market centers and SEC-registered exchanges that trade NYSE- or AMEX-listed securities send their trades and quotes to a central consolidator. The Consolidated Tape System (CTS) and Consolidated Quotation System (CQS) data streams are produced to be distributed worldwide. CQS calculates a National Best Bid and Offer (NBBO) based on the price, size, and time priority scheme for every quote message received from a market center. If the quote is a NASDAQ market maker quote, CQS also calculates a NASDAQ BBO.

We collect additional market data from multiple sources: stock price and volume data, used to estimate liquidity and volatility measures, are downloaded from CRSP; accounting and financial figures are taken from COMPUSTAT and used as control variables for firm's fundamental attributes and in capital structure analysis. Institutional investors' holdings and identities are pulled from Thomson-Reuters' Institutional (13F) Holdings data. Analyst coverage and forecasts are taken from IBES Thomson-Reuters.

The control variables in our regressions are summarized in Table 1: total assets, EBITDA, intangible assets, total liability, revenue, and leverage ratio (debt to asset).

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3.2. Measuring Liquidity and Price Efficiency

Prior studies have compared the execution costs in the NYSE and NASDAQ markets primarily on the basis of their quoted bid-ask spreads. Along with this measure, we also use two other measures for execution quality: effective and realized spreads. If trades can occur inside the spread, a better measure of execution costs is effective spread, which is based on trade price. Effective spread is given by the signed difference of the trade price minus the midpoint of the consolidated best bid or offer at the time of order receipt (Equation (1)). Effective spread is a standard measure in microstructure, and it captures the overall cost of executing a trade from the viewpoint of a trader submitting a marketable order. While the quoted spread reflects disseminated quotes, the effective spread reflects trades at implicit quotes not publicly disseminated. Therefore, it is an appropriate proxy for the liquidity component of the execution cost. To control for the magnitude of the price, relative effective spread is preferred to be used:

$$Relative_Effective_Spread_k = \frac{2D_k(P_k - M_k)}{P_k} \tag{1}$$

where D_k is a dummy variable that equals +1 if the kth trade is a buy and -1 if the kth trade is a sell, P_k is the price of the kth trade, and M_k is the midpoint of the NBBO quotes assigned to the kth trade.

Realized spread is the signed difference between the execution price and the midpoint of the consolidated quote five minutes after the trade (Equation (2)). Realized spreads capture liquidity suppliers' revenue net of losses to better-informed traders. Therefore it is a proxy for price discovery and informational efficiency. To control for the magnitude of the price, relative realized spread is preferred to be used:

$$Relative_Realized_Spread_k = \frac{2D_k(P_k - M_{k+5})}{P_k} \tag{2}$$

where M_{k+5} is the prevailing midpoint five minutes after the midpoint M_k. In the fol-

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lowing sections, we use effective spread as the measure for liquidity and the realized spread for price efficiency.

3.3. Descriptive Statistics

Table 1 presents descriptive statistics of the market cap and fundamentals of the firms in the two panels, switchers to NYSE and to NASDAQ. There are 72 firms in our sample group that switched to NYSE and 110 firms that switched to NASDAQ, during the period of 2009-2017. The average numbers indicate that, unconditionally, the switchers to NASDAQ are larger firms than the switchers to NYSE, during the period of 2009-2017. However, the average leverage ratio (debt to assets) is about 55% for both groups. The difference in revenue indicates that, on average, firms that switch to NASDAQ have more revenues than those moving in the opposite direction. On the one hand, this result is in line with the average market cap that suggested a larger size firms go to NASDAQ. On the other hand, it is somewhat unexpected, as it is against the common perception that firms move to NYSE after having grown and thus qualified for a larger, more reputable market. The minimum values for all the variables are positive, except for EBITDA, suggesting that some firms reported operational losses at the time of the switch.

[Insert Table 1 here]

Table (1) also presents the descriptive statistics for the market characteristics of the firms in the two panels on the day before the switch. The mean values of daily return, volatility (daily return standard deviation), bid-ask spread, and maximum price over minimum price ($MaxOvMin$) in the two panels are not statistically different. The distributions of these variables are slightly skewed towards the left in both panels. Standard deviation and min values indicate that dispersion was a little higher in those who left NYSE. Volume (daily number of traded shares) in those who left NYSE is larger, but the difference is not significant. These figures indicate that the two groups of switchers exhibit strongly comparable market

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characteristics at the time of switch.

Tables (2) shows the summary statistics of dark pool activities on the day before the switch. *Dark_Share* is the ratio of dark trades to all trades in a day, and, unconditionally, is about 40%. The average trade volumes indicate that, on average, the volume of trade in the lit market (exchange) is 70% higher than in the dark market. The summary statistics' numbers are cross-checked with O'Hara and Ye (2011), Foucault, Pagano, Roell, and Röell (2013).

[Insert Table 2 here]

4. Empirical Analysis and Results

4.1. Switch to NASDAQ and Cost of Execution

Despite the convergence in fees and microstructures, the preferencing agreements are more common in NASDAQ than in NYSE, and this inefficiency and friction in matching with the best available price leads to higher execution cost in NASDAQ. To test hypothesis (1), we measure the change in different market characteristics, including measures for volatility and liquidity, upon the switch to NASDAQ, and control for several co-variates. The univariate and multivariate results are reported in the appendix. Since the results could be potentially driven by factors that we failed to identify and control for in the estimation, we resolve the potential endogeneity problem by identifying a set of control firms, matching each switching company with a non-switching peer. Crucially, we identify matching observations from the population of similar firms in the same market as the switching firm is about to leave and on pre-switch financial characteristics. We use PSM (Propensity Score Matching) to find a benchmark stock for each of the switching (treatment) stocks, based on market cap, industry, and liquidity, prior to the switch time on pre-switch market. We managed to match 42 pairs for switchers to NASDAQ, and 52 pairs for switchers to NYSE. Unmatched

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firms are dropped from the diff-in-diff analysis. We estimate the following diff-in-diff model:

$$Z_{i,t} = \alpha + \beta_1 prepost_{i,t} + \beta_2 Treat_{i,t} + \beta_3 prepost_{i,t} \times Treat_{i,t} + \Phi Controls + \gamma_i + \delta_t + \epsilon_{i,t}$$

where *prepost* is a dummy variable which is 0 before switch and 1 after switch, and the vector of controls consists of $MarkCap$, act, $EBITDA$, and $Leverage$. γ and δ are respectively year and industry fixed effects. *treated* is a dummy variable, 1 if the firm is treated (switched), and 0 if the firm is a control.

Table (3) shows the results of the tests on market characteristics for both NYSE to NASDAQ and NASDAQ to NYSE switch directions. We compute daily returns as $DailyReturn = ln(\frac{P_t}{P_{t-1}})$. We measure volatility by the daily rolling standard deviation based on the last 30-day returns ($RtrnSTD$) and the ratio of the daily maximum and minimum trading prices ($MaxOvMin$). Similarly, we measure liquidity using the relative quoted bid-ask spread, $BidAskSpread = \frac{ask-bid}{bid}$, and the Amihud illiquidity measure[1]. In Panel (1), the two coefficients of the interaction terms in models (4) and (5) show that the execution cost increases when a firm switches to NASDAQ. The ratio of the daily maximum and minimum trading prices ($MaxOvMin$) increases by 21%, and the relative quoted bid-ask spread, $BidAskSpread$, increases by 13%. Our results show that liquidity is negatively affected by the switch from NYSE to NASDAQ, with relatively large magnitude.

[Insert Table 3 here]

[1] Amihud (2002) presents the liquidity measure for each stock i as the average ratio of the daily return to the (dollar) trading volume on that day, over D_i days: $ILLIQ_i = \frac{1}{D_i} \sum \frac{|R_{id}|}{VOL_{id}}$

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4.2. Shift of Trades From Lit to Dark Market: An Evidence for the Pecking Order Hypothesis

Dark markets have a lower cost of trade and higher price efficiency compared to exchange (lit) markets. In dark pools, the trade request waits until an opportunity happens with in the range of NBBO as oppose to market orders in the lit markets. Therefore, we conjecture that, following the switch to NASDAQ and the increase in the execution cost, a significant volume moves from lit markets to dark pools and the volume of trades in dark pools increases, alleviating the high cost in NASDAQ. The model specification is (4).

$$V_{i,t}^{Dark} = \alpha + \beta_1 prepost_{i,t} + \beta_2 Large_{i,t} + \beta_3 prepost_{i,t} \times Large_{i,t} + \Phi Controls + \gamma_i + \delta_t + \epsilon_{i,t} \quad (3)$$

$V_{i,t}^{Dark}$ is the trade volume in the dark markets. Table (4) shows the change in dark trade volume for switches from NYSE to NASDAQ during the period 2009-2017. Model (1) shows that after switching to NASDAQ, about 113,000 shares of a switching stock move to dark daily. Following Cox (2019), we conjecture that the fragmentation of the trades is more pronounced when the firm size is large. Therefore, we interact the dummy variable *prepost* of switch with *Large*, a dummy variable for firm size. *Large* is 1 if the total asset of the firm is above 1100 million dollars, the median of the size for the firms who switched to NASDAQ, and 0 otherwise. The results show an increase in dark trades for firms with a total asset value of above 1100 million dollars after the switch to NASDAQ, confirming our hypothesis. In Model (2), the coefficient of the interaction term indicates that, upon the switch, 418,000 shares change the trade venue from lit to dark. This number translates to about 50% increase in daily trade volume for a stock in the dark pools. To show that this number of trades are siphoned from the lit to dark markets, Models (3) and (4) measure the change in the ratio of dark trades to all trades (dark and lit). Model (3) shows a 0.8% increase in the dark trade ratio. Model (4) shows that this increase is 2% for the large firms.

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[Insert Table 4 here]

Shifting the trades from lit to dark increases the liquidity in the dark market. Table (5) presents the effects of switch to NASDAQ on the liquidity of dark venues. Model (1) shows an increase in the relative effective spread, and Model (2) shows that the effect changes its direction for the large firms. While the liquidity increases for large firms, Models (3) and (4) do not show any significant change in the relative realized spread.

[Insert Table 5 here]

4.3. Effect of Shifting the Trades to Dark on the Lit Market Quality

According to Zhu (2014), uninformed traders are more sensitive to trade cost and prefer to trade in dark markets when the trade cost in exchanges is high. Therefore, we conjecture that the responses of informed and uninformed traders are not the same when a cost shock arrives. That said, the uninformed traders are more likely to depart exchanges and go dark, leaving the exchange market more populated by the informed ones. Once the relative number of informed traders increases, price discovery improves. This improvement in price discovery comes at the expense of higher illiquidity and execution cost in the lit market. To test this hypothesis, we estimate the following cross-sectional model:

$$Y_{i,t} = \alpha + \beta_1 prepost_{i,t} + \beta_2 Large_{i,t} + \beta_3 prepost_{i,t} \times Large_{i,t} + \Phi Controls + \gamma_i + \delta_t + \epsilon_{i,t} \quad (4)$$

$Y_{i,t}$ is the relative effective/realized spreads. Table (6) presents the effects on lit markets. Models (3) and (4) show that realized spread decreases upon the switch to NASDAQ, showing an improvement in informational efficiency and price discovery.

[Insert Table 6 here]

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4.4. Switch to NASDAQ and Market Reaction

The transfer of uninformed traders to dark would have an implication: lower adverse selection cost in the lit market, although the aggregate quoted bid-ask would increase, due to lower dealer/broker competition in NASDAQ. To see how market reacts to this trade-off between cost and market efficiency, we run an event study around the switch announcement date. We estimate a standard market-adjusted model with a 180 days estimation window of (-210;-30). Abnormal returns and Cumulative Abnormal Returns (CAR) are computed from 5 days before the announcement day up to 10 days after the announcement. Figure (2) presents cross-sectional averages of the cumulative abnormal market returns around the switch announcement date at 95% confidence interval. CARs are insignificantly different from zero in either direction, confirming that investors see the switch as a trade-off with no significant value creation.

[Insert Figure 2 here]

5. Why Do Firms Switch to NASDAQ?

Finding the reasons that motivate the switch in either direction has become even more difficult. In recent years, NYSE and NASDAQ have converged in their type of microstructure and quality, as well as requirements and fees for listing. Weston (2000) shows that market reforms on NASDAQ mitigated the differences in spreads between the NASDAQ and NYSE. Although NYSE used to be, by a considerable margin, a more prestigious and more disciplined market, which was only open to "mature" companies, than NASDAQ, nowadays the services and qualities, in terms of price discovery and liquidity, are almost the same for both. Among many reforms on the two exchanges, one can refer to the two that paved the way for the listed firms to switch from one to the other: first, NYSE became a hybrid market in early 2007 due to the increasing popularity of electronic trading. Second, in July 2007, the

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SEC approved a measure under which companies with a three-letter symbol on one exchange could migrate to a different exchange and maintain that symbol. This regulatory change is important because, from the investor's point of view, changing ticker involves changing all the databases and making the new data compatible with the previous one. Also, it would require an extra effort for marketing and informing (mostly non-institutional) investors about the new ticker. Among the similarities that the two markets have converged to, fees and direct costs are of importance to study. Corwin and Harris (2001) show that although the direct issue costs of listing in NYSE is higher, total issue costs (including underpricing) do not differ across NYSE and NASDAQ and are unlikely to affect the listing decision. Macey and O'Hara (2002) study the economics of listing fees on NYSE. The immediate listing cost, which consists of a fixed (entry or listing fee) and a variable (annual fee linked to the number of shares outstanding) portion, is not significantly different between the two exchanges. In more detail, NASDAQ has three different tiers (NASDAQ (2017a)): The NASDAQ Global Select Market, The NASDAQ Global Market, and The NASDAQ Capital Market. Governance requirements are the same across all three tiers, while financial requirements are most stringent in The NASDAQ Global Select Market. The entry fee ranges from 75,000$ to 250,000$ depending on the number of total shares outstanding. Also, the annual fee has a range from 42,000$ to 155,000$. In NYSE, the entry (listing) fee is in the range of 100,000$ to 150,000$, and the annual fee ranges from 30,000$ to 85,000$ (NYSE (2017)). While these numbers apply to common shares, the rates change for other securities, such as preferred shares and structured products. Overall, the difference in fees between the two exchanges is relatively small and not large enough to compensate for the direct and indirect switching costs.

Despite the similarities and convergence, we see migrations of firms from one venue to the other, in almost the same number. There have been a number of papers and research to study the benefits of switch in each direction and try to explain the managers' decision. Some papers analyzed the switches regardless of their directions. Webb (1999) analyzes the

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timing for switch, and shows that managers switch immediately after a good performance, and that performance tends to become poorer after the switch. Some papers propose differences between asset pricing models in the two exchanges. Goyal, Pérignon, and Villa (2008) demonstrate the absence of complete similarity in return factors between NYSE and NASDAQ. They show that, although common factors exist, there are exchange-specific factors that systematically affect performance. Kaul, Mehrotra, and Stefanescu (2006) show that when firms move from NASDAQ to NYSE, the stock price process begins to co-move with the NYSE index, and its co-movement with the NASDAQ index declines. This result is at odds with fundamentals-based co-movement and is in line with the habitat-based co-movement of Barberis, Shleifer, and Wurgler (2005). It supports that moving to NYSE generally has positive effects on stock pricing and efficiency. One important implication is that moving to NASDAQ should conversely generate adverse effects. This is, however, at odds with the recent evidence. Cheng and Stunda (2015) propose that NASDAQ listing has a positive effect on short term returns, while NYSE listed firms have a better long term investment payoff.

The extant literature on the listing choice focuses primarily on companies switching to NYSE, and covers the period prior to NYSE's transition to a hybrid market. Most of the papers provide evidence of a positive market reaction and a liquidity and volatility improvement favoring the switch from NASDAQ to NYSE. Jain and Kim (2006) show that firms that switch to NYSE experience positive cumulative abnormal returns due to the increase in their visibility. Atkins and Dyl (1997) show that trading volume decreases for those companies switching their listing from NASDAQ to NYSE, but this can be attributed to the absence of dealers in NYSE at the time of the study. In an effort to explain the positive effects of switching to NYSE, Rao, Davis, and Ward (2000) identified embeddedness, social identity, and ties to in-group and out-group affiliation as common factors that positively affected switching firms. In particular, they show that firms with growth potential are usually listed on NASDAQ. Frieder and Suk (2006) show that firms listed on NYSE are more likely

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to disclose earning forecasts. Lin and Liao (2015) study the manager's financial reporting behavior in relation to exchange switching and show that moving to a more disciplined exchange mitigates the manager's incentive to inflate the financial reporting. Kedia and Panchapagesan (2011) show that cost of capital decreases by moving to NYSE due to better visibility and liquidity, and firms take this action before raising debt or equity or before an acquisition. Baker, Powell, and Weaver (1998) show that, despite the belief that switching from NASDAQ to NYSE increases media visibility, in the short term, the change in the visibility is not significant. However, in the long term, after switching, small- and medium-size firms gain more visibility. They show that the increase in media coverage is more related to rapid earnings growth before the switch than it is to the change in listing.

In this final section, we complete the studies by Tse and Devos (2004) and Dang et al. (2018) and examine other rival hypotheses to further explain the switch to NASDAQ. Similar to Dang et al. (2018) and in order to gain insight about the drivers behind the decision to switch, we looked at the reasons stated by the firms at the time of the announcement of the switch. We manually downloaded all top managers' interviews, firms' press releases, and statements on the company websites published at the time of the announcement or in the immediate vicinity (+10 days) from Factiva. Given the heterogeneity of the narratives, we have asked two research assistants to classify the motivations into thirteen categories and then compared the classification results. The two classifications were extremely coherent, and the few discrepancies have been resolved by the authors. Figure (3) reports the results of this exercise. The most commonly reported reasons for a listing switch include visibility, peer affiliation, appeal to institutional investors, cost reduction, improvement in liquidity and volatility, and, finally, the use of technological services.

[Insert Figure 3 here]

In the following, we loosely follow the managerial statements to identify and test another set of possible reasons to switch to NASDAQ.

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Industry Effect (Peer Group Benefit). Barberis et al. (2005) suggest that once firms are perceived to be in the same business environment, their stocks are more likely to co-move. Figure (4) reports an industry breakdown of switchers. The breakdown shows that switching firms are distributed across a large variety of industries with no clear difference across the two directions of switch, thus, ruling out the hypothesis that firms belong to a specific market or seek to list in a market more densely populated by peers.

[Insert Figure 4 here]

Institutional Investors. Table (7) shows the results of the cross-sectional tests on institutional investors for both directions. We adopt two measures of institutional investor ownership: the number of institutional investors (*NumInvestors*) and the concentration of ownership (*Concent10*) calculated as the ratio of the number of shares of the top 10 biggest shareholders to the total number of shares held by institutional investors. We find insignificant decreases in institutional investors' concentration.

Market Visibility. Table (7), presents the results of a set of multivariate regressions testing the change in analyst coverage before and after the switch. We compute the number of analysts announcing a forecast on the firm's target price if the announcement is made within 400 days before and after the switch. Although in both panels, the coefficients of the number of analysts are positive, indicating an increase in the number of analysts, neither of the coefficients is statistically significant. Therefore, we conclude that there is no significant change in the visibility of the firms over the course of switch.

Capital Raising (or Market Timing). We conjecture that firms' managers, facing capital needs, may choose to switch listing to attract investor attention and create a positive perception for subsequent security offerings. We test this conjecture by looking at the equity and debt issuance behavior of switching firms before and after the switch. Results in Table (7) indicate that there is no significant change in equity issuance after the switch, irrespective

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of the direction of it. Debt issuance is insignificantly different from zero after the switch to NYSE but is weakly decreasing for firms that switch to NASDAQ.

[Insert Table 7 here]

CEO Restlessness. Behavioral Finance proposes the possibility of irrational decisions by managers. One possible conjecture is that the CEOs of the switchers to NASDAQ from NYSE/AMEX have made an irrational decision. Our conjecture is that they may be "restless" managers who want to act radically to either draw attention or distract the market. One proxy to measure their maturity (opposite to restlessness) is the CEO's age. The average age of switchers, as well as the average age of all the firms who were eligible to switch to NASDAQ over the 2010-2017 period, is 56, which is not too small. Therefore we can (loosely) reject the hypothesis that the managers of the switchers, on average, are younger than those of typical firms. In fact, they are not young at all. Figure (5) shows the distribution of CEO's age for the switchers to NASDAQ and for all the firms on NYSE who were eligible to move to NASDAQ.

[Insert Figure 5 here]

To conclude, we do not find any strong rival explanation against our trade-off hypothesis. Therefore, we propose price-time trade-off as another significant reason, along with other proposed explanations in the literature, for the rationale of the switch to NASDAQ.

6. Conclusion

In this paper, we showed a substitute for the trade-off between trading in auction and dealer markets. We showed that once the firms change their listing from an auction market (NYSE) to a dealer market (NASDAQ), the traders adjust their trade venue in other segments of the market, namely dark and lit venues. We showed that upon the switch from NYSE to NASDAQ, trades shift from lit (exchange) markets to dark (off-exchange) ones.

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As a consequence, the liquidity in the dark market increases and in the lit market decreases. However, the price discovery in the lit market improves at the expense of lower liquidity.

We propose that managers choose to switch to NASDAQ to increase price discovery at the expense of execution cost. While our other alternative hypotheses do not show sound empirical supports, the trade-off hypothesis can partially explain the rationale of the switch, along with other previous explanations in the literature.

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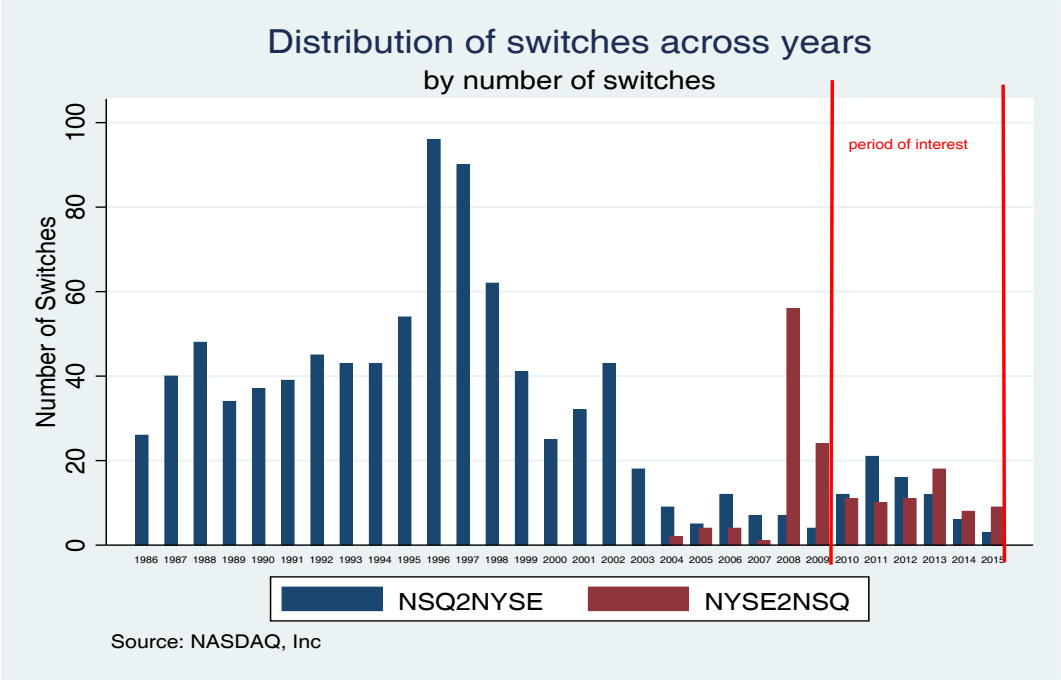
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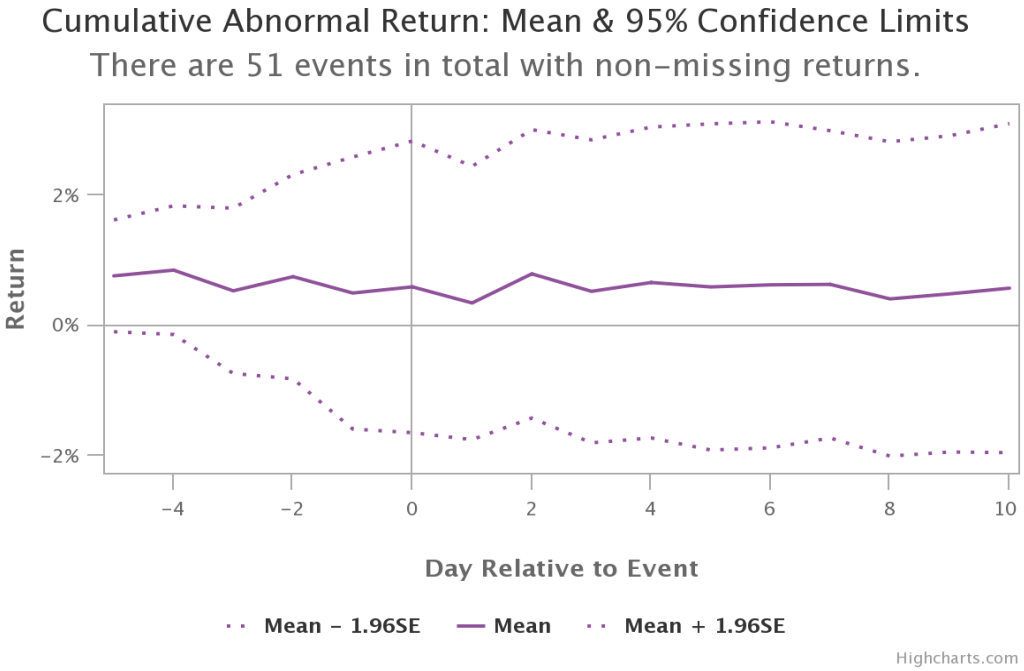
Fig. 1. Distribution of Switches between NYSE and NASDAQ during the Period of 1988-2016



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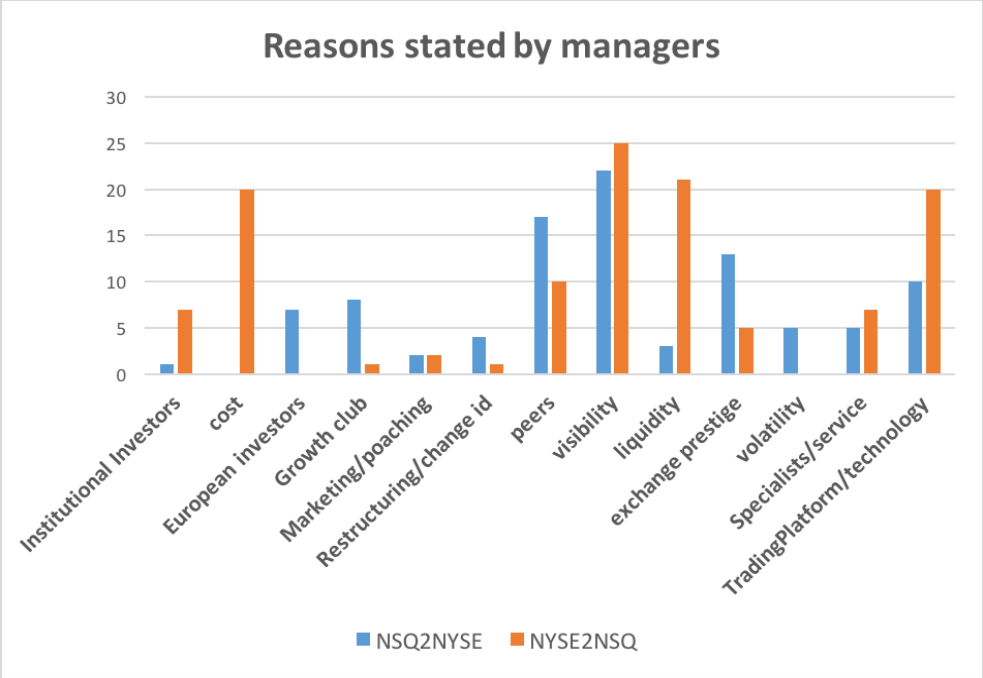
Fig. 2. Change in Switching Shares' Abnormal Return; CAR(-5;10)

CAR (Cumulative Abnormal Return) is reported. Risk model is Market-Adjusted model. Event window starts 5 days before the event (switch) and ends 10 days after. 100 trading days are used to estimate the expected return and residual return variance. 50 trading days is considered between the end of estimation window and the beginning of the event window.



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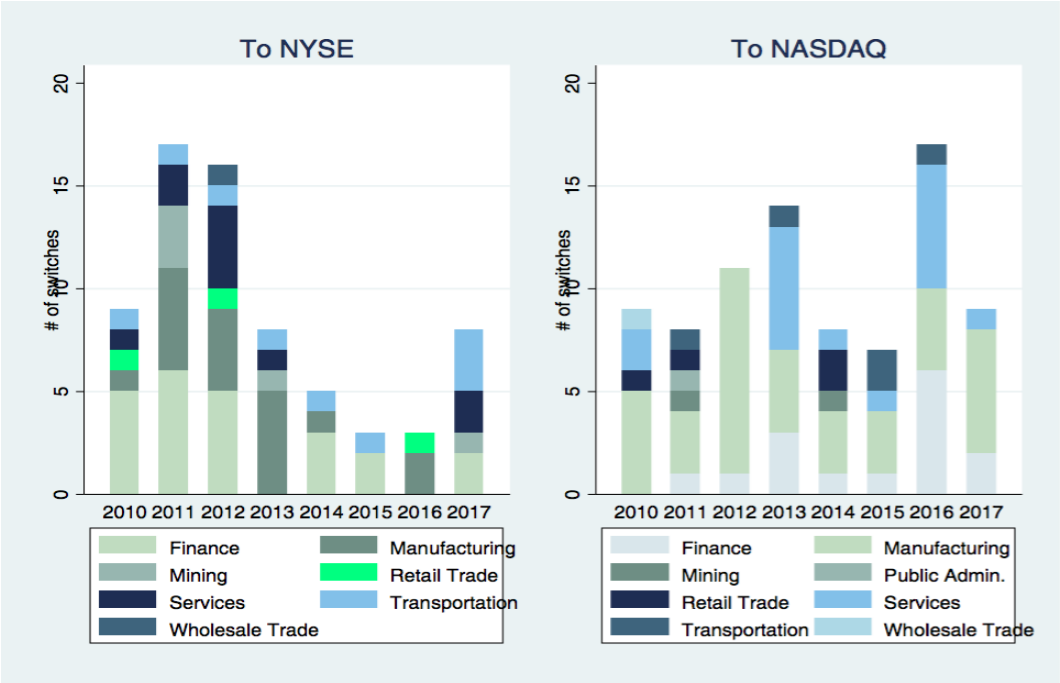
Fig. 3. Managers' Stated Motivations for the Change of Listing



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Fig. 4. Distribution of Switchers across Different Industries

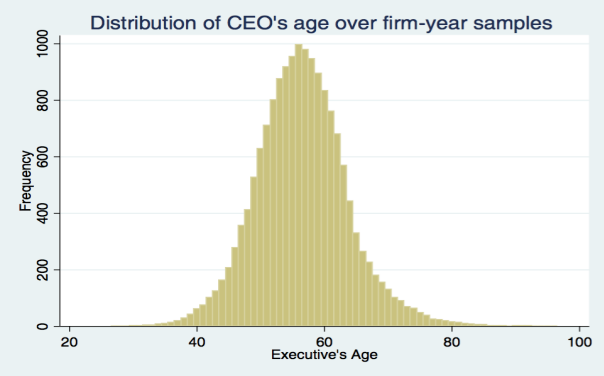
This figure reports the distribution of switchers by industry during 2010-2015, in both directions. For both panels the vertical axis is the number of switchers



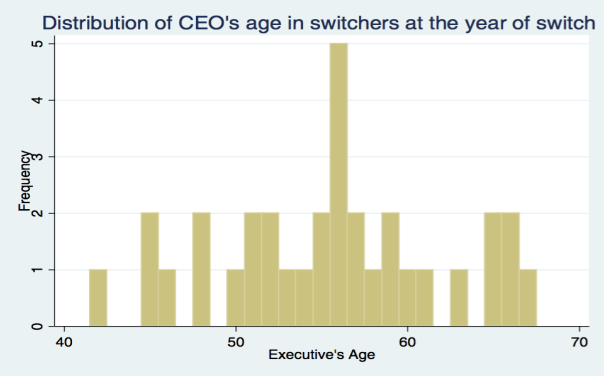
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Fig. 5. CEO's Age Effect

The two histograms show the distribution of CEO's age in all Execucomp firms, and in firms that switched to NASDAQ. For the latter, the age at the time of switch is observed. The mean of either group is 56.



(a) All firms



(b) Switchers to NASDAQ

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Table 1: Descriptive Statistics of the Switchers

This table presents descriptive statistics for switchers to NASDAQ and to NYSE in the period 2009-2017 at the day of switch. The data is from COMPUSTAT and CRSP databases. There are 110 firms switched to NASDAQ during 2009-2017, and 72 switched to NYSE. $MarkCap$ is firm's market cap, in million dollars. act is total current assets. at is total assets. $Intgb$ is intangible assets. lt is total liabilities. $Leverage$ is firm's debt to asset ratio. $Price$ is the daily last price. $DailyReturn$ is share's daily return, $log(P_t/P_{t-1})$. $RtrnSTD$ is the i-day standard deviation of $DailyReturn$. $BidAskSpread$ is the relative difference of Closing Ask and Closing Bid prices, $(P_{ask} - P_{bid})/P_{bid}$. $MaxOvMin$ is the ratio of maximum and minimum trading prices in the day, P_{askHi}/P_{bidLo}. $Volume$ is the number of shares traded daily. Significance at 10%, 5% and 1% level is denoted by by *, ** and *** respectively.

	N	mean	stdev	min	max
Panel 1) To NASDAQ:					
$MarkCap$	110	7,590	22,200	26	169,000
act	71	2,336	5,018	6	31,027
at	110	11,450	32,361	15	253,941
$EBITDA$	81	941	2,130	-289	13,189
$Intgb$	110	2,366	7,325	0	48,353
lt	110	8,612	28,710	2	240,918
$Revenue$	110	5,130*	12,322	0	76,392
$Leverage$	110	0.554	0.312	0.094	1
$Price$(\$)	110	21.442	27.844	1.312	129.745
$DailyReturn$	110	.002	.053	-.192	.337
$RtrnSTD$	78	0.035	0.046	0.006	0.358
$BidAskSpread$	110	0.004	0.006	0	0.029
$MaxOvMin$	110	1.039	0.044	1.004	1.302
$Volume(\#shares)(DarkandLit)$	110	2,043,559	7,058,595	600	58,600,000
Panel 2) To NYSE:					
$MarkCap$	72	4,715	18,400	27.6	148,232
act	57	2,409	7,816	8	48,138
at	72	7,377	17,479	13	92,568
$EBITDA$	67	638	2,315	-45	17,834
$Intgb$	69	1,620	5,772	0	35,789
lt	72	5,176	12,938	2.804	86,342
$Revenue$	72	2,328	5,744	0.050	38,275
$Leverage$	72	0.591	0.243	0.072	1
$Price$(\$)	72	26.083	23.261	1.553	113.140
$DailyReturn$	72	0	0.027	-0.114	0.083
$RtrnSTD$	72	0.024	0.029	0.008	0.235
$BidAskSpread$	72	0.003	0.006	0	0.042
$MaxOvMin$	72	1.035	0.027	1	1.142
$Volume(\#shares)(DarkandLit)$	72	3,840,224	19,300,000	517	157,000,000

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Table 2: Descriptive Statistics of the Dark Trades

This table presents descriptive statistics for the trade volumes, daily relative effective spread (D_rlt_Eff), and relative realized spread (D_rlt_Relz) in dark trades, before and after the switch to NASDAQ. The sample consists of 110 firms switched to NASDAQ during 2009-2017. $DailyTrade$ is daily trades (shares). $Dark_Share$ is the ratio of dark trades to all trades in a day.

	N	mean	median	stdev	min	max
BEFORE SWITCH:						
$DailyTrade$ (Dark and Lit)	5,691	2,306,016	519,122	4,954,340	300	7.69e+07
$DailyTrade$ (Dark Trades)	5,691	816,880	178,565	2,088,935	29	3.67e+07
$DailyTrade$ (Lit Trades)	5,691	1,489,135	321,250	3,017,149	84	4.38e+07
$Dark_Share$	5,691	.368	.346	.151	.002	.989
$D_rlt_Eff(\%)$	4,916	.210	.103	.344	.002	10.237
$D_rlt_Relz(\%)$	4,916	.321	.094	6.362	-157.581	310.773
AFTER SWITCH:						
$DailyTrade$ (Dark and Lit)	6,704	2,196,705	358,078	5,693,757	7	1.29e+08
$DailyTrade$ (Dark Trades)	6,704	818,068	131,411	2,413,060	3	5.03e+07
$DailyTrade$ (Lit Trades)	6,704	1,378,636	212871	3427634	0	7.87e+07
$Dark_Share$	6,704	.384	.369	.152	.004	1
$D_rlt_Eff(\%)$	5,718	.282	.135	.662	.005	30.942
$D_rlt_Relz(\%)$	5,718	.236	.107	16.068	-608.486	875.471

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Table 3: Switch from/to NASDAQ and Change of Trade Cost

This table reports the diff-in-diff regression of liquidity/volatility on the switch. *prepost* is a dummy variable: 0 before the switch and 1 after the switch. Treat is dummy equal to 1 if the firm is treated (switched). *DailyReturn* is share's daily return. $RtrnSTD$ is the 1-day standard deviation of the daily return. *BidAskSpread* is the relative difference of Closing Ask and Closing Bid prices. $MaxOvMin$ is the ratio of maximum and minimum trading prices in the day. $Volume$ is the number of shares traded daily. *AmihudIliquidity* is Amihud illiquidity measure, $abs(DailyRetur)/(Volume * Price)$. We control for all the con-founders mentioned in table 1, industry fixed effect, and year fixed effect. All dependent variables are in percentage. Standard errors in parentheses. Significance at 10%, 5% and 1% level is denoted by by *, ** and *** respectively.

Panel 1) To NASDAQ:

	DailyReturn	$RtrnSTD$	*AmihudIliquidity*	$MaxOvMin$	*BidAskSpread*
	(1)	(2)	(3)	(4)	(5)
prepost	-0.102	-0.645 ***	0.000	-0.601 ***	-0.012 *
	(0.001)	(0.000)	(0.000)	(0.001)	(0.000)
Treat	0.014	-0.163	-0.045 ***	0.104	0.002 ***
	(0.001)	(0.001)	(0.000)	(0.001)	(0.000)
prepost $\times Treat$	-0.037	-0.096	0.000	0.209 **	0.136 ***
	(0.001)	(0.001)	(0.000)	(0.001)	(0.000)
Controls	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES
Observations	13459	13494	13457	13539	13539
Adjusted R^2	0.002	0.208	0.024	0.176	0.302

Panel 2) To NYSE:

	DailyReturn	$RtrnSTD$	*AmihudIliquidity*	$MaxOvMin$	*BidAskSpread*
	(1)	(2)	(3)	(4)	(5)
prepost	-0.042	-0.204 ***	0.004 ***	-0.157 *	-0.017
	(0.001)	(0.000)	(0.000)	(0.001)	(0.000)
Treat	0.192	0.135	0.024	0.104 ***	-0.242 ***
	(0.001)	(0.000)	(0.000)	(0.001)	(0.000)
prepost $\times Treat$	-0.208 *	-0.136 **	0.001 ***	-0.457 ***	-0.079 **
	(0.001)	(0.001)	(0.000)	(0.001)	(0.000)
Controls	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES
Observations	11832	11879	11800	11894	11894
Adjusted R^2	0.001	0.281	0.125	0.170	0.352

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Table 4: Switch to NASDAQ and Change in Dark Trade Volume

This table reports the multivariate regression of daily dark trade volume ($Dark_Trade$) (in thousand) on the binary variable of switch from NYSE to NASDAQ. $Dark_Share$ is the ratio of dark trades to the sum of dark and lit trades in a day. $Large$ is a dummy variable which is 1 if the total asset of the firm is above 1100 million dollars, the median of the size for the firms who switched to NASDAQ, and 0 otherwise. 110 firms (switchers) are in our sample during the period of 2009-2017. Window size is 90 days. Cushion-size is 10 days. We control for all the confounding variables mentioned in table 1 and year fixed effect. Standard errors in parentheses. Significance at 10%, 5% and 1% level is denoted by by *, ** and *** respectively.

	$Dark_Trade$	$Dark_Trade$	$Dark_Share$	$Dark_Share$
	(1)	(2)	(3)	(4)
prepost	113.324 ***	-83.663 ***	0.008 ***	-0.002
	(27.791)	(17.241)	(0.002)	(0.004)
prepost × Large		418.316 ***		0.021 ***
		(58.204)		(0.004)
Large		334.612 ***		-0.142
		(70.019)		(0.158)
log_at	-93.041 ***	-43.589 ***	-0.003	-0.000
	(12.277)	(12.486)	(0.032)	(0.032)
Controls	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Constant	428.535 ***	359.922 ***	0.615 ***	0.611 ***
	(55.192)	(63.186)	(0.173)	(0.173)
Observations	12413	12413	12395	12395
R^2	0.594	0.596	0.366	0.367

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Table 5: Switch to NASDAQ and Dark Market Liquidity

This table reports the multivariate regression of daily relative effective spread (D_rlt_Eff) and relative realized spread (D_rlt_Relz) in dark trades on the binary variable of switch from NYSE to NASDAQ. *Large* is a dummy variable which is 1 if the total asset of the firm is above 1100 million dollars, the median of the size for the firms who switched to NASDAQ, and 0 otherwise. All daily measures are averaged volume-weighted in a day. Relative effective spread and relative realized spread are in percentage. 110 firms (switchers) are in our sample during the period of 2009-2017. Window-size is 200 days. Cushion-size is 10 days. Standard errors in parentheses. Significance at 10%, 5% and 1% level is denoted by by *, ** and *** respectively.

	D_rlt_Eff	D_rlt_Eff	D_rlt_Relz	D_rlt_Relz
	(1)	(2)	(3)	(4)
prepost	0.048 ***	0.068 ***	-0.094	-0.282
	(0.007)	(0.013)	(0.099)	(0.196)
prepost × Large		-0.037 **		0.343
		(0.015)		(0.212)
Large		-2.563 ***		-4.344 ***
		(0.430)		(0.962)
log_at	0.498 ***	0.490 ***	0.736 ***	0.809 ***
	(0.084)	(0.086)	(0.169)	(0.187)
Controls	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Constant	-2.252 ***	-2.230 ***	-3.337 ***	-3.544 ***
	(0.455)	(0.462)	(0.893)	(0.945)
Observations	10634	10634	10634	10634
R^2	0.204	0.204	0.002	0.003

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Table 6: Switch to NASDAQ and Lit Market Quality

This table reports the multivariate regression of daily relative effective spread (D_rlt_Eff) and relative realized spread (D_rlt_Relz) in lit (exchange) trades on the binary variable of switch from NYSE to NASDAQ. *Large* is a dummy variable which is 1 if the total asset of the firm is above 1100 million dollars, the median of the size for the firms who switched to NASDAQ, and 0 otherwise. All daily measures are averaged volume-weighted in a day. Relative effective spread and relative realized spread are in percentage. 110 firms (switchers) are in our sample during the period of 2009-2017. Window-size is 200 days. Cushion-size is 10 days. Standard errors in parentheses. Significance at 10%, 5% and 1% level is denoted by by *, ** and *** respectively.

	D_rlt_Eff	D_rlt_Eff	D_rlt_Relz	D_rlt_Relz
	(1)	(2)	(3)	(4)
prepost	-0.019 **	0.018	-0.085 ***	-0.123 **
	(0.009)	(0.016)	(0.032)	(0.056)
prepost × Large		-0.072 ***		0.072
		(0.018)		(0.066)
Large		-3.164 ***		-2.371 ***
		(0.789)		(0.829)
log_at	0.609 ***	0.598 ***	0.429 ***	0.440 ***
	(0.157)	(0.157)	(0.165)	(0.165)
Controls	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Constant	-2.905 ***	-2.887 ***	-1.952 **	-1.971 **
	(0.847)	(0.845)	(0.887)	(0.886)
Observations	11110	11110	11110	11110
R^2	0.276	0.277	0.004	0.004

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Table 7: Switching the Listing and Alternative Reasons

In this table, we report the effect of switch on several alternative variables. There are 110 firms switched to NASDAQ and 72 firms switched to NYSE during 2009-2017. *prepost* is a dummy variable: 0 before the switch and 1 after the switch. $NumInvestors$ is the number of investors holding the firm's shares. $Concentration10$ is the ratio of number of shares held by top 10 investors to the total number of shares outstanding. $NumAnalyst$ is the number of analysts covering the firm. NEI is quarterly ratio of net equity (total stock holders' equity minus retained earnings) over total assets. NDI is quarterly ratio of total long-term debt over total assets. We control for all the confounding variables mentioned in table 1, industry fixed effect and year fixed effect. Window-size is 180 days. Cushion-size is 30 days. Standard errors in parentheses. Significance at 10%, 5% and 1% level is denoted by by *, ** and *** respectively. Variables with (%) label are in percentage.

	$NumInvestors$	$Concentration10(\%)$	$NumAnalyst$	$NEI(\%)$	$NDI(\%)$
	(1)	(2)	(3)	(4)	(5)
Panel 1) TO NASDAQ					
prepost	-16.032	-0.902	0.371	-1.613	-1.152*
	(25.260)	(0.018)	(1.935)	(0.010)	(0.006)
Panel 2) TO NYSE					
prepost	-50.731	0.040	0.333	0.106	0.160
	(36.215)	(0.020)	(36.215)	(0.006)	(0.005)

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Appendix

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Table A1: Two-sample t-test for market effects

This table reports the univariate ttest of the market liquidity/volatility before and after switch. The estimated value is Mean(AfterSwitch)-Mean(BeforeSwitch). Standard errors in parentheses. Significance at 10%, 5% and 1% level is denoted by by *, ** and *** respectively.

	Panel A (To NYSE)	Panel B (To NSQ)
DailyReturn	-0.001 ***	-0.001
	(0.000)	(0.001)
N	15025	14188
RtrnSTD	-0.001 ***	-0.005 ***
	(0.000)	(0.001)
N	15043	14255
AmihudIL	-0.000 **	-7.88e-08
	(0.000)	(0.000)
N	14933	14179
LogVolume	0.043	-0.232 ***
	(0.031)	(0.044)
N	15044	14276
MaxOvMin	-0.003 ***	-0.003 ***
	(0.000)	(0.001)
N	15061	14350
BidAskSpread	-0.002 ***	0.001 ***
	(0.000)	(0.000)
N	15061	14349

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Table A2: Multivariate analysis of market effects

This table reports the multivariate regression of liquidity and volatility. We control for all the cofounding variables mentioned in table 1, industry fixed effect and year fixed effect. Window-size is 180 days. Cushion-size is 30 days. Standard errors in parentheses. Significance at 10%, 5% and 1% level is denoted by by *, ** and *** respectively.

Panel A (To NYSE)

	DailyReturn	RtrnSTD	AmihudIL	LogVolume	MaxOvMin	BidAskSpread
prepost	-0.002 **	-0.004 ***	0.000 ***	-0.160 ***	-0.006 ***	-0.001 ***
	(0.001)	(0.000)	(0.000)	(0.023)	(0.001)	(0.000)
Controls	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	7904	7922	7868	7917	7932	7932
Adjusted R^2	0.000	0.281	0.123	0.804	0.194	0.341

Panel B (To NSQ)

	DailyReturn	RtrnSTD	AmihudIL	LogVolume	MaxOvMin	BidAskSpread
prepost	-0.001	-0.007 ***	-3.56e-08	-0.340 ***	-0.006 ***	0.001 ***
	(0.001)	(0.001)	(0.000)	(0.029)	(0.001)	(0.000)
Controls	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	9566	9580	9566	9629	9667	9667
Adjusted R^2	0.001	0.204	0.011	0.761	0.190	0.345

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Table A3: Diff-in-diff analysis of market effects

This table reports the diff-in-diff regression of liquidity/volatility. We control for all the confounders mentioned in table 1, industry fixed effect and year fixed effect. *interact* is the interaction variable. Standard errors in parentheses. Significance at 10%, 5% and 1% level is denoted by by *, ** and *** respectively.

Panel A (To NYSE)

	DailyReturn	RtrnSTD	AmihudIL	LogVolume	MaxOvMin	BidAskSpread
prepost	-0.000	-0.002 ***	0.000 ***	-0.045	-0.001 *	-0.000
	(0.001)	(0.000)	(0.000)	(0.033)	(0.001)	(0.000)
Treat	0.001	0.001	0.000	0.317 ***	0.001 ***	-0.002 ***
	(0.001)	(0.000)	(0.000)	(0.032)	(0.001)	(0.000)
interact	-0.002 *	-0.001 **	0.000 ***	-0.239 ***	-0.004 ***	-0.001 **
	(0.001)	(0.001)	(0.000)	(0.042)	(0.001)	(0.000)
Controls	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	11832	11879	11800	11866	11894	11894
Adjusted R^2	0.001	0.281	0.125	0.768	0.170	0.352

Panel B (To NSQ)

	DailyReturn	RtrnSTD	AmihudIL	LogVolume	MaxOvMin	BidAskSpread
prepost	-0.001	-0.006 ***	-2.00e-08	-0.021	-0.006 ***	-0.000 *
	(0.001)	(0.000)	(0.000)	(0.032)	(0.001)	(0.000)
Treat	0.000	-0.001	-0.000 ***	1.134 ***	0.001	0.000 ***
	(0.001)	(0.001)	(0.000)	(0.033)	(0.001)	(0.000)
interact	-0.000	-0.001	-6.25e-08	-0.182 ***	0.002 **	0.001 ***
	(0.001)	(0.001)	(0.000)	(0.048)	(0.001)	(0.000)
Controls	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	13459	13494	13457	13499	13539	13539
Adjusted R^2	0.002	0.208	0.024	0.674	0.176	0.302

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Table A4: Analysis of institutional investors

This table reports the univariate, multivariate and diff-in-diff models for analyzing the effect in institutional investors. $NumInvestors$ is the number of investors holding the firm's shares. $Concentration10$ is the ratio of number of shares held by top 10 investors to the total number of shares. The observations are collected within six-quarter windows before and after the switch. The observations of three quarters before and after the switch are droppped as cushion. Standard errors in parentheses. Significance at 10%, 5% and 1% level is denoted by by *, ** and *** respectively.

Panel A (To NYSE)

	Model 1		Model 2		Model 3	
	NumInvestors	Concentration10	NumInvestors	Concentration10	NumInvestors	Concentration10
prepost	24.891	-0.037 ***	-74.573 ***	0.025	-6.962	-0.019
	(20.745)	(0.014)	(22.051)	(0.013)	(31.463)	(0.018)
Treat	-	-	-	-	79.193 ***	-0.015
					(22.988)	(0.016)
interact	-	-	-	-	-50.732	0.000
					(36.215)	(0.020)
Controls	-	-	YES	YES	YES	YES
Industry FE	-	-	YES	YES	YES	YES
Year FE	-	-	YES	YES	YES	YES
Observations	524	505	384	367	482	466
Adjusted R^2			0.772	0.696	0.813	0.634

Panel B (To NSQ)

	Model 1		Model 2		Model 3	
	NumInvestors	Concentration10	NumInvestors	Concentration10	NumInvestors	Concentration10
prepost	-5.531	0.007	27.373*	-0.031***	42.512 **	-0.031 *
	(17.695)	(0.018)	(15.198)	(0.011)	(20.161)	(0.017)
Treat	-	-	-	-	123.132***	-0.040 ***
					(19.401)	(0.014)
interact	-	-	-	-	-16.034	-0.009
					(25.260)	(0.018)
Controls	-	-	YES	YES	YES	YES
Industry FE	-	-	YES	YES	YES	YES
Year FE	-	-	YES	YES	YES	YES
Observations	494	464	425	396	658	644
Adjusted R^2			0.645	0.797	0.492	0.527

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Table A5: Change in Analysts Coverage

This table reports multivariate analysis of analysts coverage. *NumAnalyst* is the number of analysts covering the firm. The observations are collected within a 400-day window, before and after switch. The data of two months, centered at the time of switch, are dropped. Standard errors in parentheses. Significance at 10%, 5% and 1% level is denoted by by *, ** and *** respectively.

	Panel A (To NYSE)	Panel B (To NSQ)
	NumAnalyst	NumAnalyst
prepostCov	0.333	0.371
	(1.192)	(1.935)
Assets - Total	0.000 ***	0.000 *
	(0.000)	(0.000)
Year FE	YES	YES
Industry FE	YES	YES
Observations	72	70
Adjusted R^2	0.487	0.314

Table A6: Univariate analysis of capital raising

This table reports univariate analysis of equity and debt issuance. NEI is quarterly ratio of net equity (total stock holders' equity minus retained earnings) over total assets. NDI is quarterly ratio of total long-term debt over total assets. The observations are collected within a 400-day window before and after switch. The data of two months, centered at the time of switch are dropped. It is the Standard errors in parentheses. Significance at 10%, 5% and 1% level is denoted by by *, ** and *** respectively.

	Panel A (To NYSE)		Panel B (To NSQ)	
	NEI	NDI	NEI	NDI
prepostSw	0.001	0.001	-0.016	-.011*
	(0.006)	(0.005)	(0.010)	(0.006)
Constant	0.016 ***	0.008**	0.035 ***	0.012**
	(0.004)	(0.004)	(0.008)	(0.005)
Observations	457	526	444	478
Adjusted R^2	-0.002	-0.002	0.004	0.005

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